

Soham Shirke

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EDUCATION

- **Thadomal Shahani Engineering College** Bandra, Mumbai
Bachelor of Engineering in Computer Engineering; CGPA: 8.82 Aug. 2024 – June 2028
- **Pace Junior Science College** Dadar, Mumbai
Higher Secondary Certificate (Science); Marks: 85.67% June 2022 – June 2024

TECHNICAL PROJECTS

- **SourceMind** | *Next.js, TypeScript, tRPC, Zod, Neon Postgres (pgvector), Inngest* [GitHub](#) | [Live](#)
 - **RAG Knowledge Platform:** Built a serverless, event-driven multimodal RAG and knowledge synthesis platform that ingests PDFs, YouTube videos, scraped web pages, and images for semantic querying.
 - **Event-Driven Architecture:** Designed a decoupled background job pipeline using Inngest to handle long-running scraping and rendering tasks, offloading them from Next.js serverless API routes with rate-limit-aware queuing.
 - **Vector Search:** Implemented semantic search using Voyage AI embeddings stored in Neon PostgreSQL with pgvector, accelerated via HNSW indexing and cosine distance queries.
 - **Type-Safe Full Stack:** Used tRPC and Zod for end-to-end type-safe communication and schema validation across frontend and backend, catching contract mismatches at compile time.
 - **Generative Synthesis:** Enabled synthesis of interactive learning assets including mindmaps, quizzes, podcasts, and infographics using integrated generative AI models via the Vercel AI SDK.
- **CodeCook** | *Node.js, Express, PostgreSQL, Redis, BullMQ, Zod* [GitHub](#) | [API Docs](#)
 - **Competitive Coding Platform API:** Built a self-hosted, LeetCode-style backend API supporting authentication, coding problems, playlists, leaderboards, and an admin panel.
 - **Async Code Execution:** Implemented an asynchronous submission pipeline using BullMQ and Redis to queue code execution jobs, offloading them to a remote Piston execution engine and writing verdicts back to PostgreSQL.
 - **Authentication & Authorization:** Implemented JWT-based authentication with HTTP-only cookies, refresh tokens, and role-based access control for admin-only operations.
 - **Validation & Documentation:** Used Zod for request/response schema validation and set up auto-generated interactive API documentation using Scalar, with containerized local infrastructure (PostgreSQL, Redis) via Docker Compose.
- **Google Doc Clone** | *TypeScript, Next.js, Clerk, Convex, Liveblocks, Zustand* [GitHub](#)
 - **React-Based Editor UI:** Built a Google Docs-style rich text editor interface in React with a dynamic, context-aware toolbar and clean, component-driven layout.
 - **Authentication:** Integrated Clerk for secure user sign-in and session handling across the app.
 - **Responsive Interactions:** Built smooth document switching and toolbar interactions backed by centralized state management with Zustand, keeping the interface fast and responsive across views.

TECHNICAL SKILLS

- **Languages:** C, Python, HTML, CSS, JavaScript, TypeScript
- **Frameworks:** React, Next.js, Express.js, FastAPI
- **Databases:** MongoDB, PostgreSQL, Redis
- **Validation:** Zod
- **Tools:** Git, GitHub, Docker, Swagger API

NOTABLE ACHIEVEMENTS

- **TSEC CodeCell Coding Challenge:** Ranked 3rd overall in a competitive 10-week coding challenge organized by TSEC CodeCell, demonstrating consistent problem-solving ability and algorithmic thinking across multiple rounds.